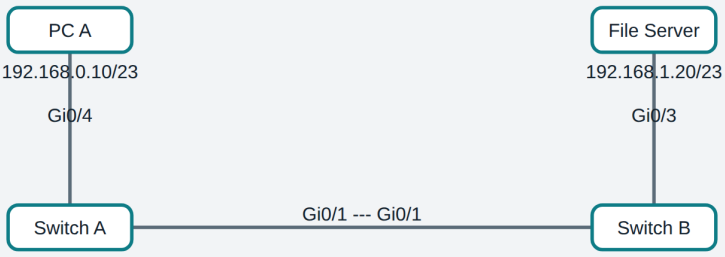
VLAN & Trunking Questions 2

5 questions | Original English questions and source answers

Question 1

Refer to the exhibit. A network administrator assumes a task to complete the connectivity between PC A and the File Server. Switch A and Switch B have been partially configured with VLANs 10, 11, 12 and 13. What is the next step in the configuration?




```
Switch A
Vlan 10,11,12,13

interface GigabitEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10-12
!
interface GigabitEthernet0/4
 switchport access vlan 13
 switchport mode access

Switch B
Vlan 10,11,12,13

interface GigabitEthernet0/1
 switchport mode trunk
!
interface GigabitEthernet0/3
 switchport access vlan 13
 switchport mode access
```

Options

A. Add a router on a stick between Switch A and Switch B allowing for inter-VLAN routing

B. Add VLAN 13 to the trunk links on Switch A and Switch B for VLAN propagation

C. Add PC A to VLAN 10 and the File Server to VLAN 11 for VLAN segmentation

D. Add PC A to the same subnet as the File Server allowing for intra-VLAN communication

Answer

B. Add VLAN 13 to the trunk links on Switch A and Switch B for VLAN propagation

Source explanation

In fact we only need to add (allow) VLAN 13 to the trunk link of Switch A. Switch B allows all VLANs by default so we don’t need to do any further configuration.

Source: CCNA 200-301 Questions.html | Question 1

Question 2

An engineer must establish a trunk link between two switches. The neighboring switch is set to trunk or desirable mode. What action should be taken?

Options

A. configure switchport trunk dynamic desirable

B. configure switchport mode dynamic desirable

C. configure switchport mode dynamic auto

D. configure switchport nonegotiate

Answer

B. configure switchport mode dynamic desirable

Source explanation

If the neighboring switch is set to trunk then we need to set our switch to desirable mode so that it can start the negotiation. In fact answer "configure switchport mode dynamic auto" is also correct according to the table below:

| | Dynamic Auto | Dynamic Desirable | Trunk | Access

| Dynamic Auto | Access | Trunk | Trunk | Access

| Dynamic Desirable | Trunk | Trunk | Trunk | Access

| Trunk | Trunk | Trunk | Trunk | Limited Connectivity

| Access | Access | Access | Limited Connectivity | Access

But “desirable” mode is a safer answer so we should choose it.

If the neighboring switch is set to trunk then we need to set our switch to desirable mode so that it can start the negotiation. In fact answer "configure switchport mode dynamic auto" is also correct according to the table below:

	Dynamic Auto	Dynamic Desirable	Trunk	Access
Dynamic Auto	Access	Trunk	Trunk	Access
Dynamic Desirable	Trunk	Trunk	Trunk	Access
Trunk	Trunk	Trunk	Trunk	Limited Connectivity
Access	Access	Access	Limited Connectivity	Access

But “desirable” mode is a safer answer so we should choose it.

Source: CCNA 200-301 Questions.html | Question 2

Question 3

An engineer requires a switch interface to actively attempt to establish a trunk link with a neighbor switch. What command must be configured?

Options

A. `switchport mode trunk`

B. `switchport mode dynamic auto`

C. `switchport nonegotiate`

D. `switchport mode dynamic desirable`

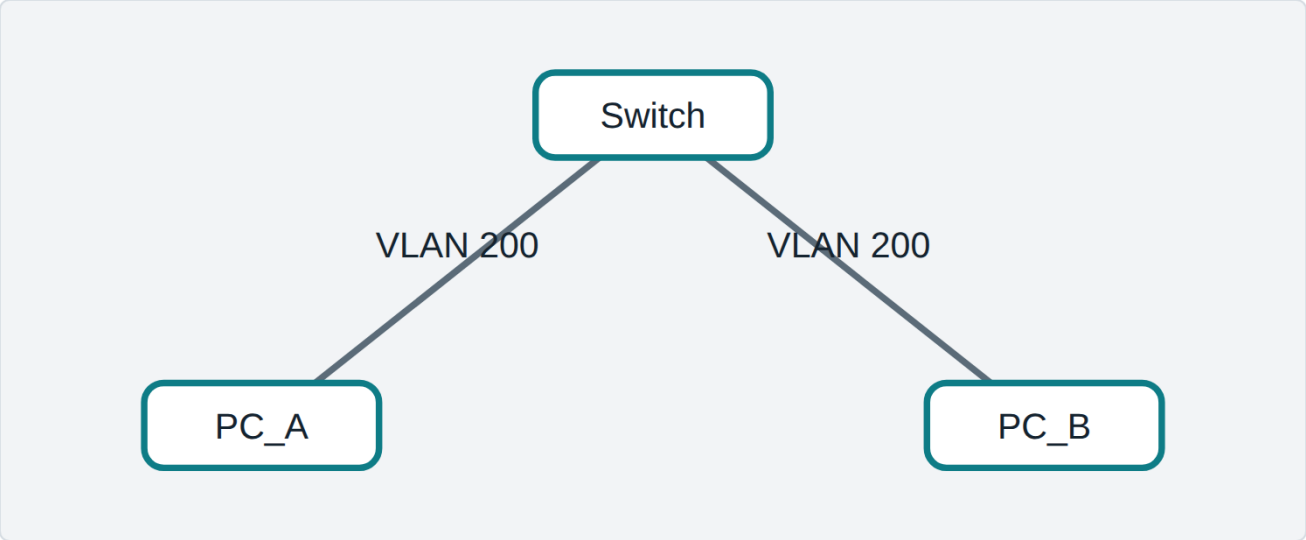
Answer

D. switchport mode dynamic desirable

Source: CCNA 200-301 Questions.html | Question 3

Question 4

Refer to the exhibit. Which outcome is expected when PC_A sends data to PC_B?



Options

A. The source MAC address is changed

B. The destination MAC address is replaced with ffff.ffff.ffff

C. The source and destination MAC addresses remain the same

D. The switch rewrites the source and destination MAC addresses with its own

Answer

C. The source and destination MAC addresses remain the same

Source explanation

Layer 2 switches do not change MAC address so PC_A will put PC_B MAC address into its destination address before sending it to the switch. PC_B will receive the frames with the source and destination MAC address intact.

Note: Only with routers, including layer-3 switches where the packets need to cross to other VLANs, will the frames be stripped and rewritten for the new network or VLAN.

Source: CCNA 200-301 Questions.html | Question 4

Question 5

An engineer must configure interswitch VLAN communication between a Cisco switch and a third-party switch. Which action should be taken?

Options

A. configure IEEE 802.1q

B. configure IEEE 802.1p

C. configure DSCP

D. configure ISL

Answer

A. configure IEEE 802.1q

Source explanation

IEEE 802.1Q is the networking standard that supports virtual LANs (VLANs) on an Ethernet network. When a frame enters the VLAN-aware portion of the network (a trunk link, for example), a VLAN ID tag is added to represent the VLAN membership of that frame. The picture below shows how VLAN tag is added and removed while going through the network.

Source: CCNA 200-301 Questions.html | Question 5